SVD

$$A = U \Sigma V^T$$

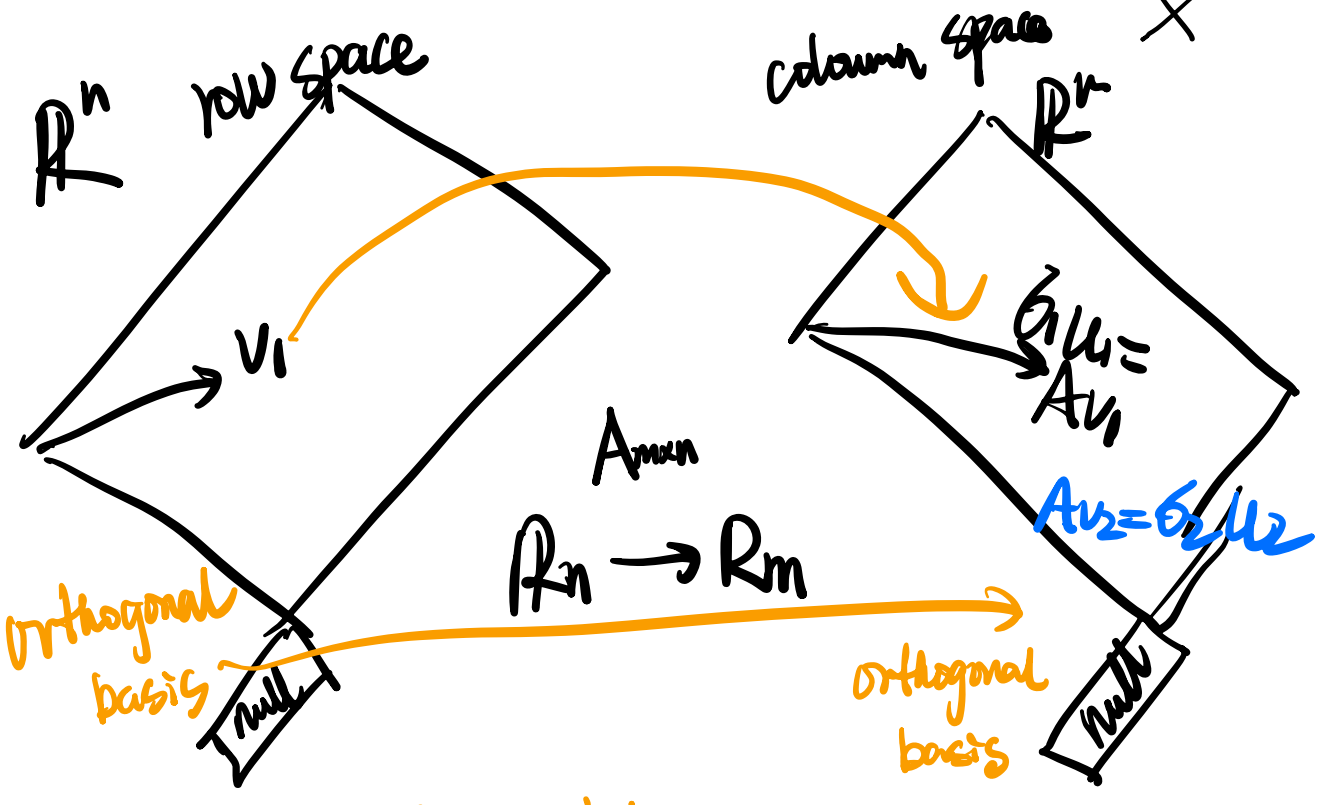
column space — orthogonal — AA^T

Σ — diagonal

row space — orthogonal — A^TA

$A = S \Lambda S^T$ — symmetric positive definite

$A = S \Lambda S^{-1}$ ×



Find A to take orthogonal basis to another orthogonal

$$A @ \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_m \\ | & | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}$$

i.e. $Av_1 = \sigma_1 u_1$

Therefore $AV = U\Sigma$

V: orthonormal, Σ: orthogonal

2	1	1
2	3	2

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

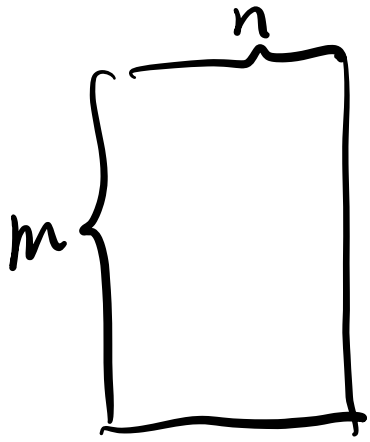
v_1, v_2 in row space, $\mathbb{R}^2$

u_1, u_2 in col space $\mathbb{R}^2$

$\sigma_1 . \sigma_2 > 0$

$\mathbb{R}^n \longrightarrow \mathbb{R}^m$

A $m \times n$



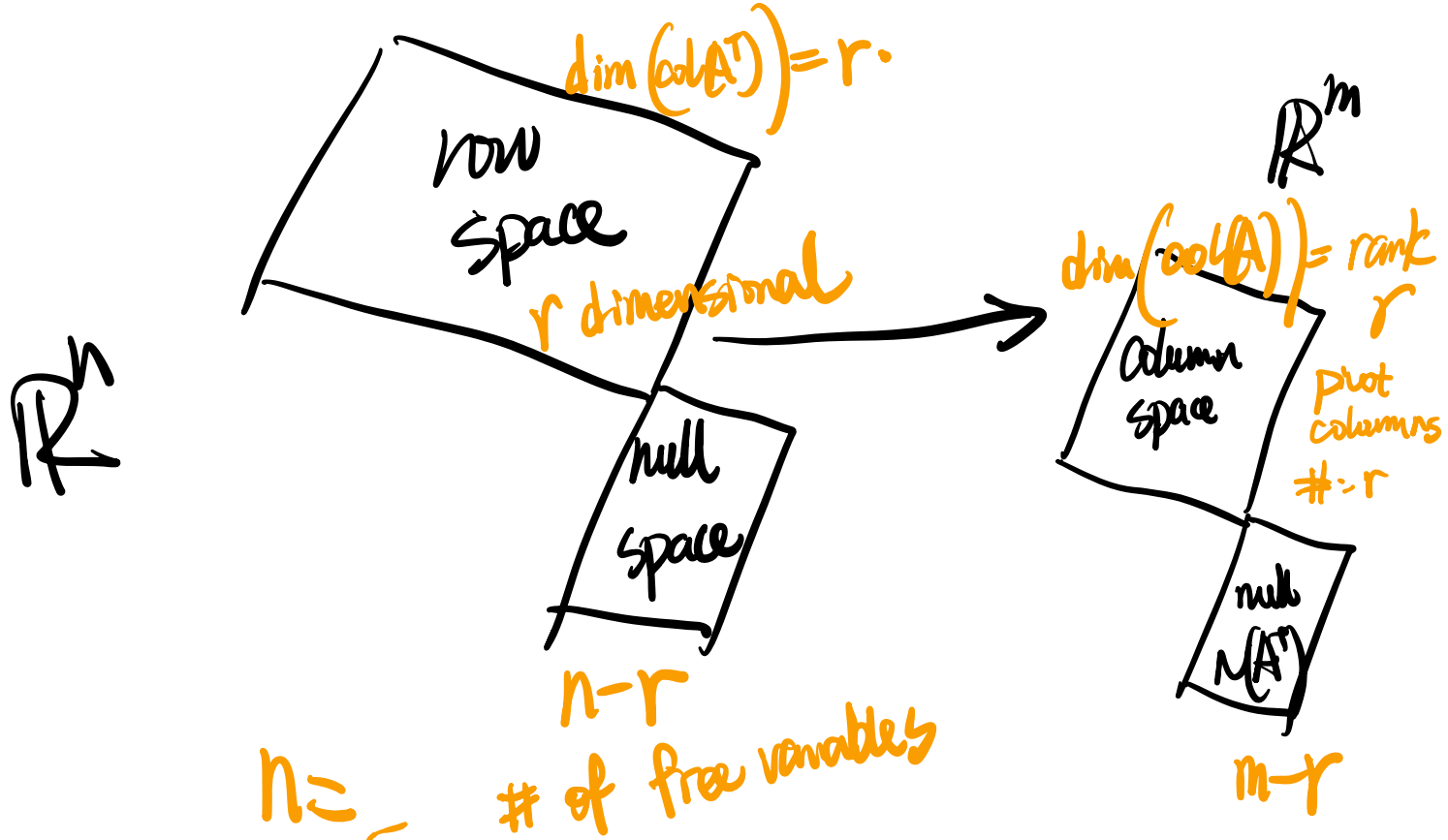
col(A) in $\mathbb{R}^m$

null(A) in $\mathbb{R}^n$

row space col(A^T) in $\mathbb{R}^n$

null (A^T) in $\mathbb{R}^m$

A: $\mathbb{R}n \to \mathbb{R}m$ $m \times n$ matrix



$n =$ # of free variables

$n = 4$ (5 columns)

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ & & & & \\ & & & & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Kernel

$$b\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$a = 0$
$d = 0$
$e = 0$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} a + 0b + 0c + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} d + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} e$$

row operation
preserve row space

but produce different column space

basis for row space
is the first rows of R

4th space $N(A^T)$ A $\mathbb{R}^n \to \mathbb{R}^m$, $m \times n$

$A^T y = \vec{0}$ A^T $\mathbb{R}^m \to n$

$$\overbrace{\begin{bmatrix} y^T \end{bmatrix}}^{m} \; {}_{m}\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}^{n} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}}_{n}$$

Left null space

r ref $[A_{m \times n} \; I_{m \times m}] \to R_{m \times n}$. Elimination

$$EA = R$$

$$E \quad A \quad R$$

$$\begin{bmatrix} -1 & 2 & 0 \\ +1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$E \qquad A \qquad R$

$I \quad 0 \quad 1$ 余

Vector space

$M_{3\times 3}(\mathbb{R})$

$A+B \qquad mA$

not AB

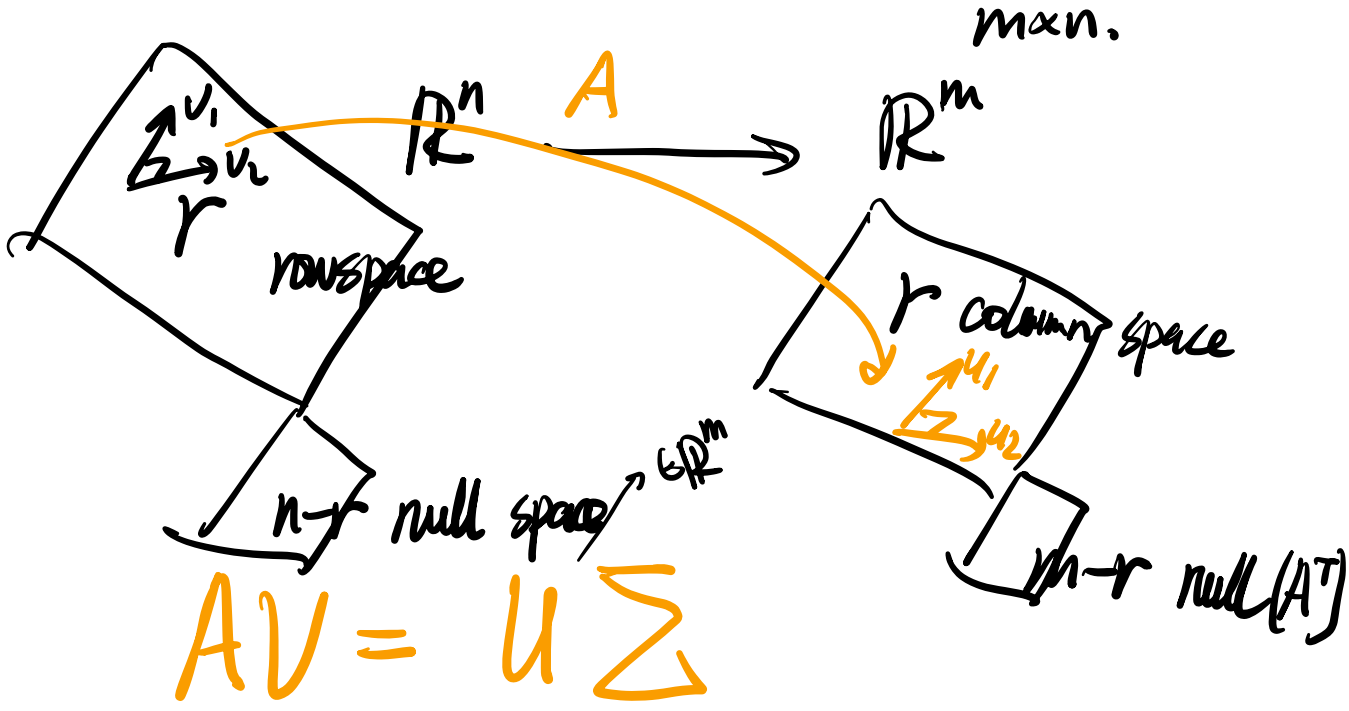
Subspace of M: All upper triangular matrices

All symmetric matrices

upper triangular $\cap$ symmetric = diagonal matrices

dimension 3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$



$$AV = U\Sigma$$

$$A_{r}\left\{\begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_r \\ | & | & & | \end{bmatrix}\right. = {}_{m}\left\{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{bmatrix}\right. \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

$$A = U\Sigma V^T$$

① $A^TA = V\Sigma^T U^T U \Sigma V^T$

$= V\Sigma^2 V^T$

σ^2 are the eigenvalue of A^TA.

② $AA^T = U\Sigma V^T V \Sigma U^T$

$= U\Sigma^2 U^T$

u: eigenvector of AA^T

$$A^TA = \begin{bmatrix} & \\ & \end{bmatrix} = V\Lambda V^T$$

$[n\times m]\,[m\times n]$ $\quad n\times n$

positive definite

$A^TAx = \lambda x$

$\begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix}$

↓ eigenvectors orthogonal.

$$AA^T = \underset{m\times n \quad n\times m}{\begin{bmatrix} & \\ & \end{bmatrix}} = U \Lambda U^T$$

$m \times m$

m eigenvalues

Looking for Av

$Av_1 = \sigma_1 u_1$

$Av_r = \sigma_r u_r$

$Av_{r+1} = 0$

$\vdots$

$Av_n = 0$

$u_1 = \frac{Av_1}{\sigma_1}$

$u_r = \frac{Av_r}{\sigma_r}$

Looking for orthonormal vectors in $v \in \mathbb{R}^n$.

and u orthogonal in $\mathbb{R}^m$

I should show that $u_1 \cdots u_r$ are orthogonal

$u_1^T u_2 = 0$ $\quad \left(\frac{Av_1}{\sigma_1}\right)^T \left(\frac{Av_2}{\sigma_2}\right)$

V is the orthonormal vectors of A^TA

$n\times m \quad m\times n \quad n\times n \quad n\times n \quad n\times n$

$A^TA = V\Sigma^2V^T$

$A: m \times n$

$= \dfrac{v_1^T A^T A v_2}{\sigma_1 \sigma_2} = \dfrac{v_1^T \sigma_2^2 v_2}{\sigma_1 \sigma_2} = 0$

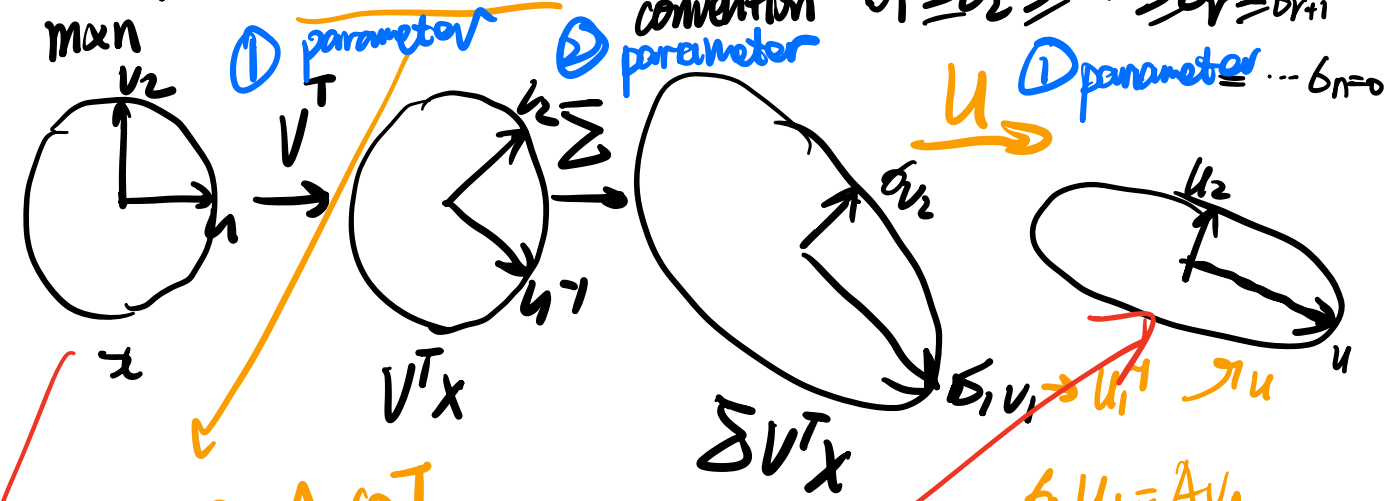
$AA^T = U\Sigma^2U$

$m\times m \quad m\times m$

$m\times n \quad m\times n \quad n\times n$

$A = U\Sigma V^T$ x

$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq \sigma_{r+1}$



$Q \Lambda Q^T$

A is squared if $A^TA = AA^T$

then $U = V$

$\rightarrow U\Sigma U^T$

A

$\sigma_1 u_1 = Av_1$

$\sigma u_1 v_1^T = A$

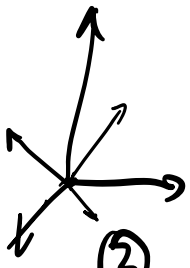
$u_1^T \sigma u_1 v_1^T = u_1^T A$

$\sigma v_1^T = u_1^T A$

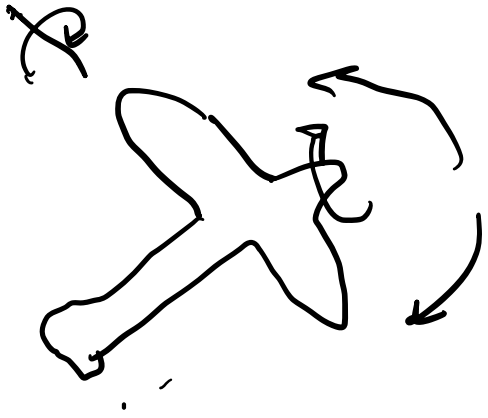
$u_1 \sigma v_1^T = u_1 u_1^T A$

$= A$

$A_{3\times 3} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$



3 parameters for rotating



$m \times n$

$$A = U \Sigma V^T$$

$m \times n$

AA^T → determinant 1

A^TA → determinant 1

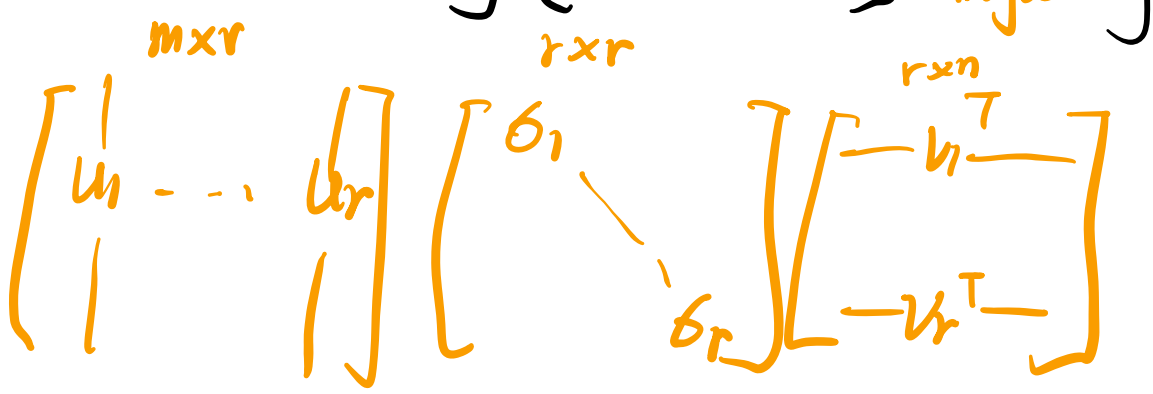
$$\sigma_1 \sigma_2 = 15 = \lambda_1 \lambda_2$$

$$\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$$

$$A = \underset{m \times n}{\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u \\ | & | & & | \end{bmatrix}} \underset{m \times r}{\underset{m \times m}{\begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ & & \sigma_r \\ 0 & & 0 \end{bmatrix}}} \underset{r \times r}{\underset{m \times n}{\begin{bmatrix} - & v_1^T & - \\ & \vdots & \\ - & v_2^T & - \end{bmatrix}}} \underset{r \times n}{\underset{n \times n}{}}$$

maybe 0

maybe 0



polar decomposition of a matrix

$re^{i\theta}$

$$A = U\Sigma V^T = \underset{\text{symmetric}}{S}Q$$

$$= U\Sigma(U^TU)V^T$$

$$= (U\Sigma U^T)UV^T$$

$$= S(UV^T)$$

$$[A] = U\Sigma V^T$$

$$= [u_1 \cdots u_r][\sigma_1 \; \sigma_r][v_1^T \; v_r^T]$$

r

$\sum_{i=1}^{r} u_i \sigma_i v_i^T$ Top principle

$A_k = \sum_{i=1}^{r} u_i \sigma_i v_i^T$ using the first k matrices are the best approximation.

If B has rank k then $\|A-B\| \geq \|A-A_k\|$

The norm of a matrix

$\|v\|_2 = \sqrt{|v_1|^2 + \cdots |v_n|^2}$ $\|v\| = |v_1| + \cdots |v_n|$

$\|A\|_2 :=$ 2 norm $:= \sigma_1$

$\|v_\infty\| = \max \|v\|_i$

$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$

Frobenius

$\|A\| = \sigma_1 + \sigma_2 + \cdots + \sigma_r$

nuclear norm

for these norms

$\|A-B\| \geq \|A-A_k\|$

$\forall B$

$$A = \begin{bmatrix} 4 & & 0 \\ & 3 & \\ & 2 & \\ 0 & & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} 4 \begin{bmatrix} 1.0 & 0.0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} 3 \begin{bmatrix} 0 & 1.0 & 0 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 3.5 & 3.5 & & \\ 3.5 & 3.5 & & \\ & & 1.5 & 1.5 \\ & & 1.5 & 1.5 \end{bmatrix}$$

$$A = U \Sigma V^T$$

Q orthonormal matrix

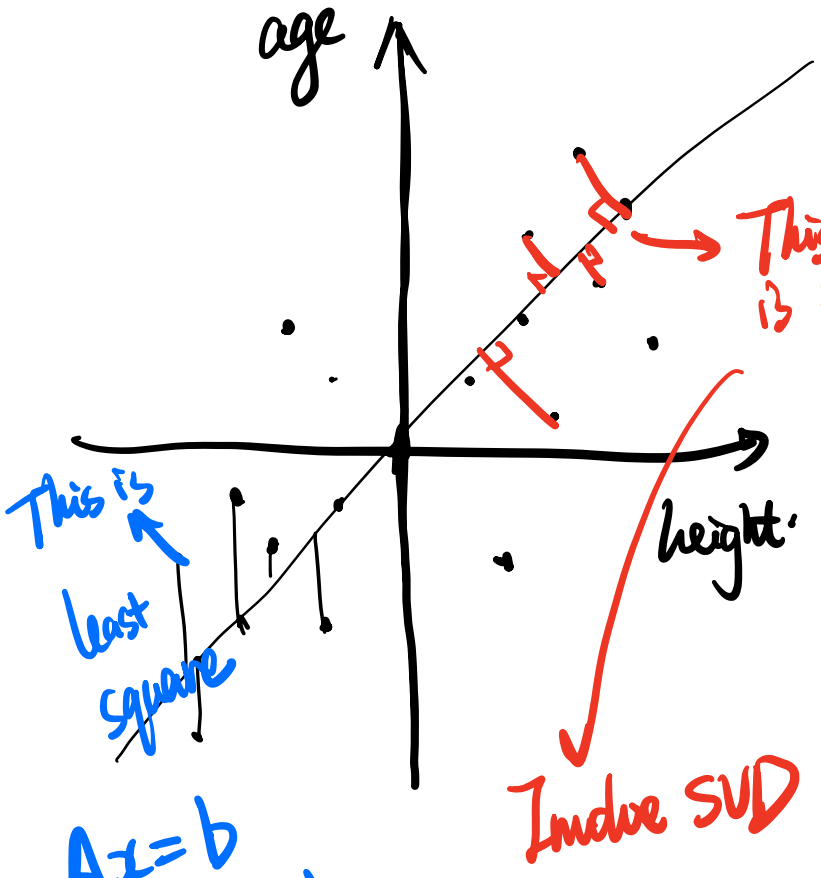
$$\|Q(v)\| = \|v\|$$

$$\|Qv\|^2 = (Qv)^T Qv$$
$$= v^T Q^T Q v$$
$$= \|v\|^2$$

$$QA = QU \Sigma V^T$$

QA singular value $(QU) \Sigma V^T$

Therefore multiply an orthogonal matrix does not change the singular value



$$A_0 = \begin{bmatrix} | & | & | & & | \\ v_1 & v_2 & v_3 & \cdots & v_n \\ | & | & | & & | \end{bmatrix} \begin{matrix} \text{height} \\ \text{age} \end{matrix}$$

$$A = A_0 - \begin{bmatrix} a_1 & a_1 & \cdots & a_1 \\ a_2 & a_2 & \cdots & a_2 \end{bmatrix}$$

covariance matrix!

$$\frac{AA^T}{N-1}$$

$Ax = b$

$A^TAx = A^Tb$

regression!

$u \in \text{col}(A) \quad v \in \text{col}(A^T)$ row space

$$= \sigma_1 u_1 v_1^T + \cdots \sigma_r u_r v_r^T$$

$$= \sigma_1 \, u_1 \, v_1^T + \sigma_2 \, u_2 \, v_2^T$$

$$= \tilde{U} \tilde{\Sigma} \tilde{V}^T$$

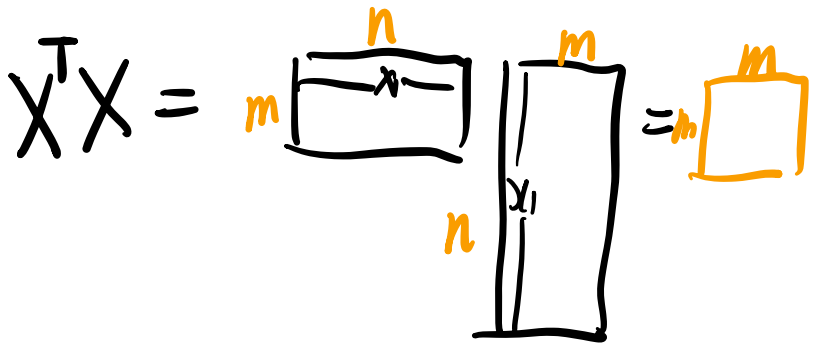
Note $U^TU = I$

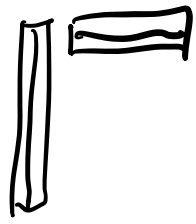
AA^T A^TA

$n >> m$

$\hat{u}^T\hat{u} = I$

$\hat{u}\hat{u}^T \neq I$





$$= \begin{bmatrix} x_1^Tx_1 & x_1^Tx_2 \cdots \\ x_2^Tx_1 & \\ \vdots & \\ x_n^Tx_1 & \end{bmatrix}$$

$XX^T = \hat{u}\Sigma^2\hat{u}^T$

$\Rightarrow XX^T\hat{u} = \hat{u}\Sigma^2$

$\hat{u}$ → eigenvectors

Left singular vectors

$Av_1 = \sigma_1 u_1$

$$X = U\Sigma V^T = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} — & u & — \\ & \vdots & \\ — & v_r & — \end{bmatrix}$$

$$= \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} — & v_1 & — \\ — & v_2 & — \end{bmatrix}$$

k

$X = n[\quad]$

$$\Sigma = \frac{X^T X}{n-1}$$

$\overline{w_1}$ unit vector in the variable space R^k

$$X\overline{w_1} = n\begin{bmatrix} -\text{sample 1}- \\ -\text{sample 2}- \end{bmatrix}_k \begin{bmatrix} | \\ \cdot \\ | \end{bmatrix}_k = \begin{bmatrix} w_1^T \text{sample 1} \\ w_2^T \text{sample 2} \end{bmatrix} \Big\} \begin{matrix} \max \\ \text{variance} \end{matrix}$$

orthonormal vector 1

Later

$$X\begin{bmatrix} | & | \\ v_1 & v_2 \\ | & | \end{bmatrix} \to \begin{bmatrix} | & | \\ Xv_1 & Xv_2 \\ | & | \end{bmatrix}$$

$$X = U\Sigma V^T$$

V : eigenvector of $X^T X$

variance $X\overline{w_1} = \dfrac{(Xw_1)^T Xw_1}{n-1}$

$$X = \begin{bmatrix} | & | \\ u_1 & u_2 \\ | & | \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} \text{weights of 1st} \\ -v_1- \\ \text{weights of 2nd} \\ -v_2- \end{bmatrix}$$

$$\max \quad w_1^T \frac{X^T X}{n-1} w$$

$\dfrac{X^T X}{n-1}$ is positive definite!

$= [(Xv_1)_1, (Xv_2)_1 \;]$

$XV = U\Sigma V^T V$

$= U\Sigma$

Claim

$\frac{(Xv_1)^T(Xv_1)}{n-1}$ is the largest!

sample 1 · v_1 | sample 1 · v_2 | sample 1 · v_r

$Xv_1 \; ; Xv_2 \cdots Xv_r$

$= \begin{bmatrix} u\sigma_1 & u_2\sigma_2 \cdots & u_r\sigma_r \end{bmatrix}$

PCA Transformed Data

PC1 PC2 PC_{rth}

gives us the transformed version of data!

XVV^T